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ORIGINAL PAPER

Three-dimensional modeling of fracture in quasi-brittle materials using plasticity and cohesive finite elements

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Abstract This paper presents a three-dimensional discrete approach based on the use of interface cohesive elements with zero thickness to model the crack formation and propagation in quasi-brittle materials. The interface constitutive model incorporates concepts of nonlinear fracture mechanics and plasticity. A Coulomb-based surface is used as yield function, and an exponential traction–separation curve as softening law. The model features a non-associated flow rule with an implicit integration scheme and is able to simulate the propagation of single and multiple cracks in mode I and mixed-mode. The proposed model was tested through extension and shear paths and applied in the simulation of experimental tests in concrete specimens available in the literature. Also, a study on the effect of the use of different bulk elements was performed. Numerical results, including peak loads and load-CMOD curves showing softening behavior, are in good agreement with the experimental data.

Keywords Finite elements · Quasi-brittle materials · Concrete cracking · Cohesive elements · Plasticity

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1 Introduction

There are several techniques to deal with the simulation of fracture in quasi-brittle materials within the context of finite elements. Those techniques can generally be classified as continuous and discontinuous approaches. On the continuous approach, the cracks are represented in a smeared manner where the openings are distributed over the finite elements. On the other hand, in the discontinuous approach, cracks are modeled as displacement discontinuities between finite elements. The use of cohesive elements (e.g. Ngo and Scordelis 1967; Nilson 1968; Bocca et al. 1990; Gerstle and Xie 1992) and the XFEM (e.g. Melenk and Babuska 1996; Wells and Sluys 2001; Belytschko et al. 1996) are examples of the discontinuous approach. One advantage of using cohesive elements over continuous models is the possibility of obtaining more realistic crack representations since it allows the visualization of crack openings due to the progressive detachment of bulk elements. Another advantage lies in the capacity to simulate and visualize a large number of cracks.

The use of cohesive elements has obtained an interest in different areas. Applications in two dimensions (2D) include areas as crack propagation in concrete (Gerstle and Xie 1992; Carol et al. 1997; Cendón et al. 2000; Yang and Chen 2005; López et al. 2008a, b; García-Álvarez et al. 2012; Lens et al. 2009; Tejchman and Bobński 2013; Benedetto et al. 2018), fiber-reinforced medium (Caggiano et al. 2012; Etse et al.

2012), geomechanics (Day and Potts 1994; Garolera et al. 2020), flow through fractures (Ng and Small 1997; Segura and Carol 2004, 2010; Carrier and Granet 2012; Nguyen et al. 2017), delamination (Geubelle and Baylor 1998; Turon et al. 2007; Corona and Reedy 2011), and general crack propagation (Molinari et al. 2007; Park et al. 2009). In turn, three-dimensional (3D) applications also include areas as crack propagation in concrete (Evangelista et al. 2013; Xu et al. 2016; Ferté et al. 2016; Jiang and Meng 2018), geomechanics (Beer 1985; Coutinho et al. 2003; Cerfontaine et al. 2015; Cui et al. 2019), delamination (Roychowdhury et al. 2002; van den Bosch et al. 2008), composites (Jin et al. 2003; Segurado and LLorca 2004; Caballero et al. 2008; Khoei et al. 2012) and general crack propagation (Ortiz and Pandolfi 1999; Pandolfi and Ortiz 2002; Nguyen 2014; Morin et al. 2013; Tabiei and Zhang 2017). Although two-dimensional simulations provide important insight into the fracture process, those are limited to plane stress and plane strain problems. Besides that, general three-dimensional state problems such as torsion cannot be analyzed using two-dimensional models.

The prediction of crack openings and crack propagation using cohesive elements can be performed by the use of the cohesive zone model (CZM) (Dugdale 1960; Barenblatt 1962; Hillerborg et al. 1976). In this model, a crack is assumed to propagate when the stress at the crack tip reaches the tensile strength. Also, the CZM includes a fracture process zone (FPZ) that allows the transfer of stresses across the discontinuity. Those stresses can be related to the crack width by using softening curves, also called traction–separation relations. Bilinear (Wittmann et al. 1988; Yang and Chen 2005; Roesler et al. 2007; Park et al. 2008; Park and Paulino 2011) and exponential (Hordijk 1992; Xu and Needleman 1994) models are examples of traction–separation curves.

The failure of a cohesive element can be predicted using the Mohr-Coulomb theory. This theory is probably one of the simplest and practical theories due to the use of few material parameters that can be obtained using widely known experimental tests. Some authors (e.g. Carol et al. 1997; Segura and Carol 2010; Carey and Ma 1999; Lens et al. 2009; García-Álvarez et al. 2012) have investigated the use of yield functions based on this theory to simulate fracture in two-dimensional conditions within the context of plasticity and cohesive elements. In general, they present a good prediction of the peak load in displacement driven crack tests. In some cases, few problems regarding the softening branch prediction can be pointed. For example, the lack of agreement with experimental data and no fully available curves. These results suggest the capability of a Coulomb-based law to predict the beginning of fracture and some difficulties related to plastic flow rules.

This work is a follow-up of a two-dimensional version (Durand and Silva 2019) where a Coulomb-based plasticity model was used. In that previous work, two different softening curves were studied (Bilinear and Hordijk's), verifying that the difference in finite element fracture simulations using these curves is negligible. Also, that study presented an analysis on the effect of different interpolation order in bulk elements, concluding that quadratic elements do not improve the results significantly. The 2D model was successfully applied in the fracture simulation of concrete specimens but restricted to plane stress conditions. On the other hand, this paper presents a 3D extension of the yield and plastic potential functions to include the additional shear stress component. As traction–separation law, the Hordijk's model is used in preference of the bilinear model. All the formulation, as well as necessary derivatives, are included in the paper. Along with the simulations, the effect of different bulk elements is also studied. The proposed model represents an alternative to 3D existing models (e.g. damage and XFEM models) due to the simplicity in terms of parameters and formulation which is given in the context of conventional plasticity. Also, it is capable of simulating, with good and fairly good agreement, a wide range of experimental tests from the literature.

Validation examples and simulations of classical experiments in 3D conditions are presented. Due to geometry and load asymmetry, most of the studied cases can only be simulated using a 3D model. Since the fracture problem is highly nonlinear, a finite element solver with automatic load/displacement stepping is used. The numerical load-CMOD curves present good and fairly good agreement with the experimental data, showing that our approach can simulate a range of cases involving mode I and mixed-mode fracture in 3D. Regarding the use of different types of bulk elements, we observed that hexahedral elements interfere with the development of cracks tending to increase the peak load or generate several cracks in regions where one crack was expected. On the other hand, the use of tetrahedral and pyramidal elements showed better

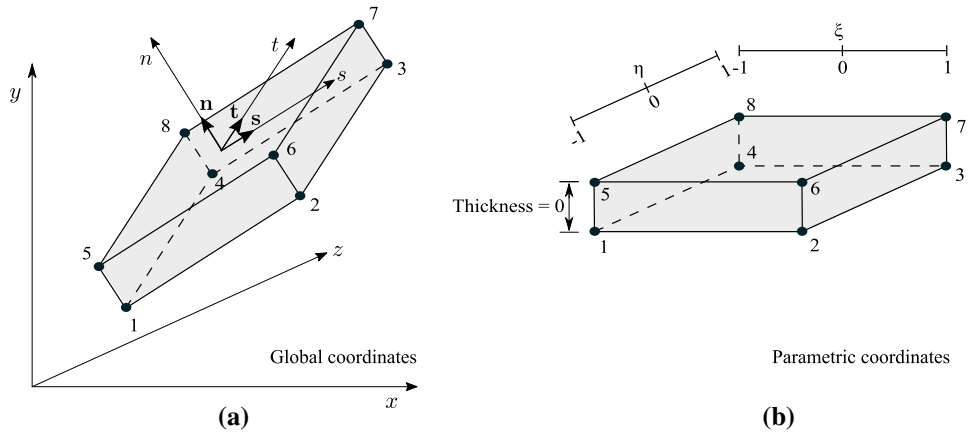


Fig. 1 Eight-node interface element: **a** global coordinate system (x, y, z) and local reference given by vectors **n**, **s** and **t**; **b** parametric coordinate system (ξ, η)

crack patterns. For problems that can be simplified to the plane stress state, wedge elements are enough to simulate the fracture process successfully.

The paper is organized as follows. Section 2 describes the basics of interface elements and the generation of interface elements in conventional meshes. Section 3 deals with the constitutive modeling for the cohesive elements. In turn, Sect. 4 deals with the numerical solution procedure. Finally, Sect. 5 presents some patch tests and application examples where laboratory experiments are successfully simulated using the proposed approach.

2 Isoparametric interface element

This section addresses the finite element formulation of an isoparametric interface element in three dimensions. Since the proposed approach requires introducing cohesive elements between bulk elements, the algorithm to generate interface elements into an existing conventional mesh is briefly described. This section also includes the procedure to compute a representative length for bulk elements linked to cohesive elements.

2.1 Finite element formulation

Finite element formulations for isoparametric interface elements are quite standard and have been addressed in previous works (Beer 1985; Plesha 1989; Hird and Kwok 1989; Roy and Dodds 2001; Jin et al. 2003). This paper presents the formulation, referring to a quadrilateral 8-node interface element; however, the equations can easily be modified to consider triangular interface elements and even higher interpolation order elements. Figure 1 presents a zero thickness interface element with contact area A. The figure at left shows the element located at the global coordinate system (x, y, z). In contrast, the figure at right shows the element in the parametric system (ξ, η).

The displacements vector for the element presented in Fig. 1 is written as

$$\mathbf{U} = \langle u_{x1},\ u_{y1},\ u_{z1},\ u_{x2},\ u_{y2},\ u_{z2},\ \cdots,\ u_{x8},\ u_{y8},\ u_{z8} \rangle. \tag{1}$$

For the given parametric coordinates (ξ, η), the relative displacements between faces are represented by vector $\mathbf{w} = \langle w_n, w_s, w_t \rangle$. This vector is calculated by interpolation as

$$\mathbf{w}(\xi,\ \eta) = \mathbf{T}(\xi,\ \eta)\mathbf{N}(\xi,\ \eta)\,\mathbf{U}, \tag{2}$$

where $\mathbf{N}(\xi, \eta)$ is a matrix based on shape functions and $\mathbf{T}(\xi, \eta)$ is a rotation matrix used to express global displacements in the local reference. The following equation shows matrix $\mathbf{T}$, where n_i, s_i and t_i are the components of vectors **n**, **s** and **t**, respectively

$$\mathbf{T} = \begin{bmatrix} n_1 & n_2 & n_3 \\ s_1 & s_2 & s_3 \\ t_1 & t_2 & t_3 \end{bmatrix} \tag{3}$$

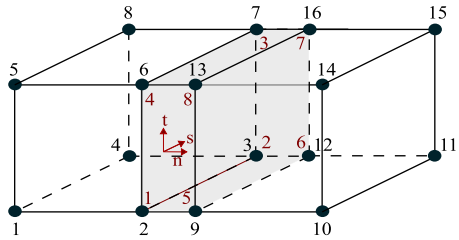


Fig. 2 Global (black) and local (red) nodal numbering for a cohesive element

The rotation matrix is easily calculated employing the Jacobian matrix of a quadrilateral element. The Jacobian matrix is defined as

$$\mathbf{J} = \begin{bmatrix} \frac{\partial x}{\partial \xi} & \frac{\partial y}{\partial \xi} & \frac{\partial z}{\partial \xi} \\ \frac{\partial x}{\partial \eta} & \frac{\partial y}{\partial \eta} & \frac{\partial z}{\partial \eta} \end{bmatrix}. \tag{4}$$

This is a non-square matrix since the mapping occurs between (x, y, z) and (ξ, η) systems. Note that each row from the Jacobian matrix represents a vector that is tangent to the face surface. This property is used to find a set of perpendicular vectors (**n**, **s** and **t**). The normal vector **n** is obtained as the unitary vector from the cross product of the Jacobian matrix lines. Vectors **s** and **t** can be chosen arbitrarily as long they are perpendicular to **n**. Vector **s** is chosen as the unitary vector from the first line of the Jacobian matrix. Finally, vector **t** is found as the unitary vector of the product $\mathbf{n} \times \mathbf{s}$. This last product is required since the rows of the Jacobian matrix are not necessarily perpendicular.

Regarding the displacement interpolation matrix **N**, it is given by

$$\mathbf{N} = \begin{bmatrix} -N_1 & 0 & 0 & \cdots & -N_4 & 0 & 0 & N_1 & 0 & 0 & \cdots & N_4 & 0 & 0 \\ 0 & -N_1 & 0 & \cdots & 0 & -N_4 & 0 & 0 & N_1 & 0 & \cdots & 0 & N_4 & 0 \\ 0 & 0 & -N_1 & \cdots & 0 & 0 & -N_4 & 0 & 0 & N_1 & \cdots & 0 & 0 & N_4 \end{bmatrix} \tag{5}$$

where N_1, N_2, N_3 and N_4 are the same shape functions as those from a quadrilateral element. Figure 2 shows an example of two bulk elements linked by a cohesive element. In this case, the interface element connectivity is given by nodes 2, 3, 7, 6, 9, 12, 16 and 13.

Subsequently, Eq. (2) can be rewritten incrementally as follows, where the symbols ξ and η were dropped for clarity. Thus,

$$\Delta \mathbf{w} = \mathbf{T}\,\mathbf{N}\,\Delta \mathbf{U}. \tag{6}$$

The stress state at each point along the interface element is represented by the vector $\boldsymbol{\sigma} = \langle \sigma_n, \tau_s, \tau_t \rangle$, where σ_n, τ_s and τ_t represent the stress components in the **n**, **s** and **t** directions, respectively. The stress increment vector $\Delta\boldsymbol{\sigma}$ is related to the relative displacements increment vector $\Delta\mathbf{w}$ by means of the constitutive matrix **D**; thus, $\Delta\boldsymbol{\sigma} = \mathbf{D}\,\Delta\mathbf{w}$.

Using a conventional finite element procedure, as the virtual work method, the stiffness matrix for an interface element is found as

$$\mathbf{K} = \int_A \mathbf{B}^T \mathbf{D} \mathbf{B}\, dA, \tag{7}$$

where $\mathbf{B} = \mathbf{TN}$. Matrix **K** is calculated using numerical integration; for this purpose, a number of integration points must be defined. For instance, in this work, quadrilateral interface elements are integrated using four integration points, and triangular interface elements are integrated using three integration points. For m integration points, the stiffness matrix is approximated as

$$\mathbf{K} = \sum_{i=1}^{m} \mathbf{B}_i^T \mathbf{D}_i \mathbf{B}_i \,\text{norm}(\mathbf{J}_i). \tag{8}$$

Since **J** is a non-square matrix, the corresponding norm is calculated as

$$\text{norm}(\mathbf{J}) = \sqrt{\begin{vmatrix} J_{11} & J_{12} \\ J_{21} & J_{22} \end{vmatrix}^2 + \begin{vmatrix} J_{12} & J_{13} \\ J_{22} & J_{23} \end{vmatrix}^2 + \begin{vmatrix} J_{13} & J_{11} \\ J_{23} & J_{21} \end{vmatrix}^2}, \tag{9}$$

where the terms J_{ij} are components of the Jacobian matrix **J**, as presented in Eq. (4).

2.2 Generation of interface elements

In the discrete approach for fracture modeling used in this work, all bulk elements are connected to their neighbors through zero-thickness cohesive elements. Therefore, a pre-processing procedure is required to insert cohesive elements into an existing conventional mesh (see e.g. Paulino et al. 2008; Nguyen 2014; Jiang and Meng 2018; Tabiei and Zhang 2017; Durand and Silva 2019). In this work, we extend the algorithm presented by Durand and Silva (2019) to 3D conditions.

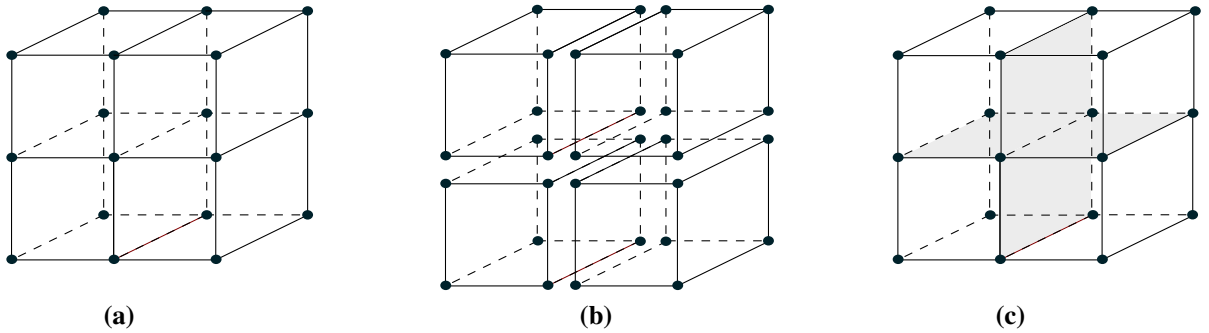


Fig. 3 Generation of interface elements: **a** inital mesh; **b** detached elements; **c** final mesh including the generated interface elements

Basically, given a conventional mesh (Fig. 3a), the algorithm starts by detaching all bulk elements from their neighbors by replacing existing connectivities with new nodes (Fig. 3b). Once all elements are detached, a hash value is found for each face of all elements. A face's hash value is calculated based on its nodal coordinates. Next, for each face pair found with the same hash value (i.e., faces at the same position but from different elements), a new interface element is generated. Finally, the mesh is updated to include all generated interface elements (Fig. 3c).

2.3 Computation of a representative length for bulk elements

In non-local constitutive modeling, the size of the corresponding finite element is required to counter mesh dependency. In the case of interface elements, the proposed formulation considers the size of the two bulk elements linked by a cohesive element. Since these sizes may be different, an individual representative length h is used for each cohesive element. Consider, for example, the meshes in Fig. 4 and assume that cohesive elements link all bulk elements. In the first case, the representative length is given by the elements' width since all bulk elements are equal. In the second case, the representative length for the cohesive element between the two shaded elements will be the average between the widths h_1 and h_2. In the last case, the value of h for the shaded elements is computed considering two prismatic elements with equivalent volumes, V_1 and V_2. This allows us to compute two equivalent widths: $h_1 = V_1/A$ and $h_2 = V_2/A$. Finally, the representative length is given, for all cases, by the average of these values. Thus,

$$h = \frac{V_1 + V_2}{2A}. \tag{10}$$

By using equivalent volumes, this procedure applies to other types of elements, as wedges, tetrahedrons and pyramids.

The representative length adopted here is used to reduce mesh dependency by countering mesh-related undesired interface displacements. Consequently, this also improves the model accuracy. The introduction of the representative length into the formulation is presented in the next section.

3 Constitutive modeling

The constitutive model for the cohesive elements is formulated using the plasticity theory and is an extension of the 2D model presented by Durand and Silva (2019). In this case, the yield function is given by a Coulomb-based surface that includes two shear stress components as

$$f(\boldsymbol{\sigma}, \sigma_{max}) = \sqrt{\tau_s^2 + \tau_t^2} + (\sigma_n - \sigma_{max}) \tan\phi = 0, \tag{11}$$

where σ_n, τ_s and τ_t are, respectively, the stress components at an integration point of a cohesive element, while σ_{max} is the internal variable, i.e. the material tensile strength, and ϕ is the material friction angle. Figure 5a shows the yield surface where $\sigma_{max}^{(0)}$ and $\tau_{max}^{(0)}$ represent, respectively, the initial tensile and shear strength of the material. Assuming a non-associated flow rule, the proposed plastic potential surface (Fig. 5b) is given by

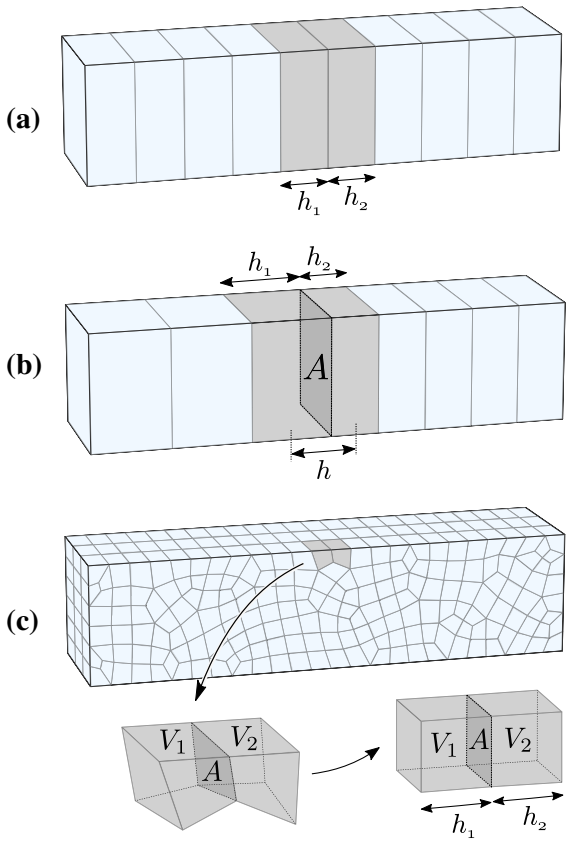


Fig. 4 Computation of the representative length of bulk elements: **a** in a mesh with equal hexahedral elements; **b** in a mesh with elements with different sizes; **c** in an unstructured mesh

$$g(\boldsymbol{\sigma}, \sigma_{max}) = \begin{cases} \tau_s^2 + \tau_t^2 - \tau_{max}^2 = 0 & \text{for} \quad \sigma_n \leq 0 \\ \sigma_n^2 \tan^2\phi + \tau_s^2 + \tau_t^2 - \tau_{max}^2 = 0 & \text{for} \quad \sigma_n > 0 \end{cases} \tag{12}$$

The internal variable σ_{max} defines the location of the yield surface. The displacement of the surface tip towards the origin aims to simulate the loss of strength due to cracking. σ_{max} is a function of the effective relative plastic displacement defined as

$$u_p = \int_0^t ||\dot{\mathbf{w}}_p||\, dt = \int_0^t \sqrt{\dot{w}_{np}^2 + \dot{w}_{sp}^2 + \dot{w}_{tp}^2}\, dt, \tag{13}$$

where $\dot{\mathbf{w}}_p = \langle \dot{w}_{np}, \dot{w}_{sp}, \dot{w}_{tp} \rangle$ is a vector containing the rates of relative displacements in the local system (n, s, t). There are several traction–separation curves available in the literature to relate σ_{max} and u_p. For example, bilinear softening curves are widely used in the literature (see, e.g. Wittmann et al. 1988; Yang and Chen 2005; Roesler et al. 2007; Park et al. 2008; Park and Paulino 2011) due to their simplicity and ability to provide accurate results in finite element analyses. Other softening models are worth mentioning, as exponential and polynomial models (see e.g. Xu and Needleman (1994), Park et al. (2009)); although they were not investigated as part of the formulation proposed in this paper. In this work, we opted to use the exponential curve proposed by Hordijk (1992) (see Fig. 6), because it provides a closed-form expression to represent the post-peak softening curve. Nevertheless, the proposed

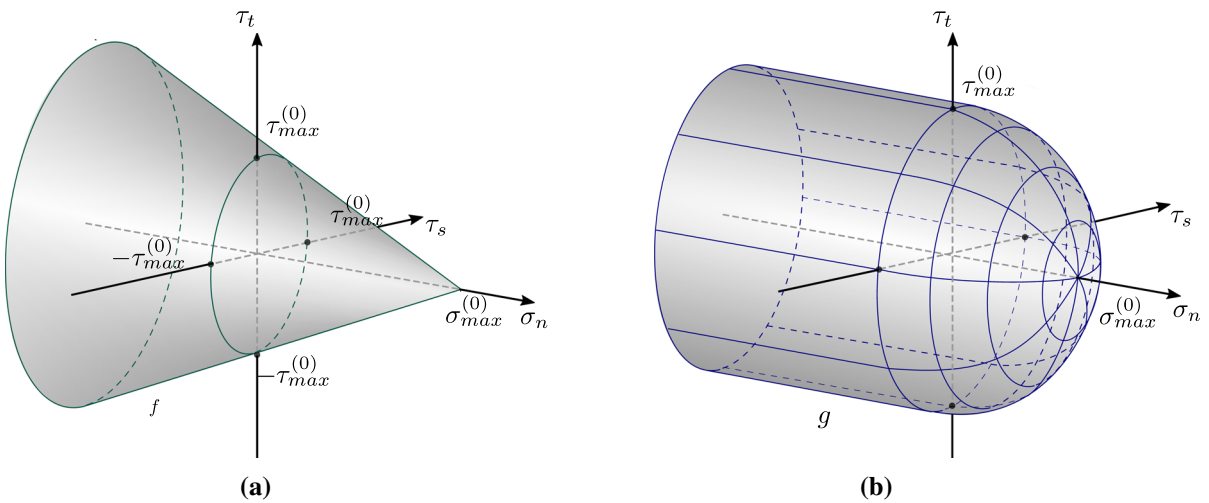


Fig. 5 Constitutive model surfaces: **a** yield function; **b** plastic potential function

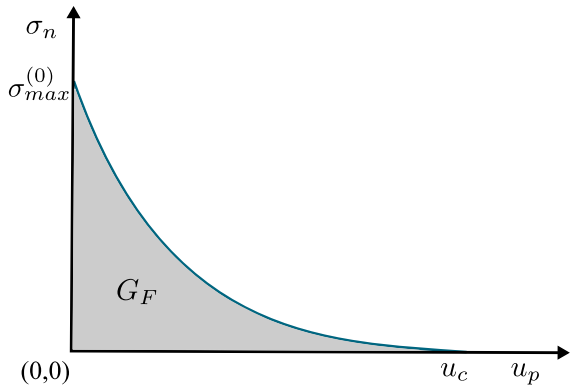


Fig. 6 Hordijk's softening curve

approach is not limited to this particular model and a different model can be used as the softening law. The Hordijk's curve depends only on two parameters, σ_{max}^0 and u_c. The term u_c represents the critical crack opening when σ_{max} reaches zero. Its value is calculated so as to achieve an area under the curve equal to the material fracture energy G_F. The Hordijk's softening curve, using the coefficients proposed by Zivaljic et al. (2014), is written as

$$\sigma_{max}(u_p) = \begin{cases} \left[\left(1+27\left(\frac{u_p}{u_c}\right)^3\right) e^{-6.93\frac{u_p}{u_c}} - 28\frac{u_p}{u_c}\, e^{-6.93}\right] \sigma_{max}^0 & \text{for} \quad u_p \le u_c \\ 0 & \text{for} \quad u_p > u_c \end{cases} \tag{14}$$

After integration of Eq. (14) from 0 to u_c and equating to G_F we get

$$u_c = \frac{G_F}{0.1947\,\sigma_{max}^0}. \tag{15}$$

Consequently, the rate of change of σ_{max} with respect to u_p is given by

$$\frac{\partial \sigma_{max}}{\partial u_p} = \begin{cases} \left(\dfrac{81\,u_p^2\, e^{\frac{-6.93\,u_p}{u_c}}}{u_c^3} - \dfrac{6.93\left(1+\frac{27\,u_p^3}{u_c^3}\right) e^{\frac{-6.93\,u_p}{u_c}}}{u_c} - \dfrac{0.02738402432}{u_c} \right) \sigma_{max}^0 & \text{for} \quad u_p \le u_c \\ 0 & \text{for} \quad u_p > u_c \end{cases} \tag{16}$$

As can be seen from the curve in Fig. 6, σ_{max} reduces when u_p increases; this causes the displacement of the yield surface, as seen in Fig. 7, allowing the simulation of loss of strength.

3.1 Constitutive matrix

The constitutive matrix $\mathbf{D}$ relates the relative displacement increment vector $\dot{\mathbf{w}}$ with the stress increment vector $\dot{\boldsymbol{\sigma}}$ at an interface integration point of the cohesive element. For the elastic regime, the constitutive matrix is given by

$$\mathbf{D}_e = \begin{bmatrix} k_n & 0 & 0 \\ 0 & k_t & 0 \\ 0 & 0 & k_t \end{bmatrix}, \tag{17}$$

where k_n and k_t are components in the normal and tangential directions. Considering the approach proposed by Lens et al. (2007) to take into account mesh effects, the elastic coefficients can be defined as

$$k_n = E\frac{\alpha}{h} \quad \text{and} \quad k_t = G\frac{\alpha}{h}, \tag{18}$$

where h is the representative length of bulk elements as described in Sect. 2.3, E is the Young's modulus, $G = E/(2(1+\nu))$ is the shear modulus, and α is a factor that controls the relative elastic displacements. The α factor can be used to estimate the maximum relative elastic displacements allowed in the normal and tangential directions according to

$$w_{ne(max)} = \frac{\sigma_{max}^0}{\alpha E} h \quad \text{and}$$
$$w_{se(max)} = w_{te(max)} = \frac{\tau_{max}^0}{\alpha G} h. \tag{19}$$

For the elasto-plastic regime, the constitutive matrix $\mathbf{D}_{ep}$ is formulated based on the conventional plasticity

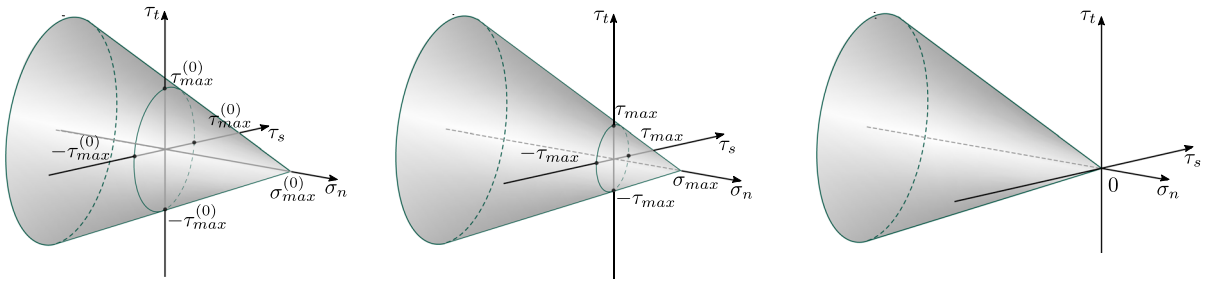


Fig. 7 Yield function evolution: **a** initial location ($u_p = 0$); **b** intermediary location ($0 < u_p < u_c$); **c** final location ($u_p \geq u_c$)

theory. According to the additive decomposition, the increment of relative displacements in a point along the interface is expressed as the sum of the elastic and plastic components:

$$\dot{\mathbf{w}} = \dot{\mathbf{w}}_e + \dot{\mathbf{w}}_p. \tag{20}$$

In turn, the increment of relative plastic displacements is calculated by the flow rule according to:

$$\dot{\mathbf{w}}_p = \dot{\lambda}\,\mathbf{r}, \tag{21}$$

where $\mathbf{r}$ is a vector normal to the potential surface g and $\dot{\lambda}$ is the plastic multiplier. Thus, the increment of the effective plastic relative displacement is calculated as

$$\dot{u}_p = ||\dot{\mathbf{w}}_p|| = \dot{\lambda}\,||\mathbf{r}||. \tag{22}$$

After multiplying Eq. (20) by $\mathbf{D}_e$, considering $\dot{\boldsymbol{\sigma}} = \mathbf{D}_e\dot{\mathbf{w}}_e$ and Eq. (21), we get the stress increment vector as

$$\dot{\boldsymbol{\sigma}} = \mathbf{D}_e\,\dot{\mathbf{w}} - \dot{\lambda}\,\mathbf{D}_e\,\mathbf{r}. \tag{23}$$

On the other hand, the consistency condition, according to Eq. (11), is given by

$$\dot{f} = \frac{\partial f}{\partial \boldsymbol{\sigma}} \cdot \dot{\boldsymbol{\sigma}} + \frac{\partial f}{\partial \sigma_{max}}\,\dot{\sigma}_{max} = 0, \tag{24}$$

which can be expanded as

$$\dot{f} = \frac{\partial f}{\partial \boldsymbol{\sigma}} \cdot (\mathbf{D}_e\,\dot{\mathbf{w}} - \dot{\lambda}\,\mathbf{D}_e\,\mathbf{r}) + \frac{\partial f}{\partial \sigma_{max}}\,\frac{\partial \sigma_{max}}{\partial u_p}\,\dot{u}_p = 0. \tag{25}$$

Substituting Eq. (22) into (25) and isolating the increment of the plastic multiplier, we get

$$\dot{\lambda} = \frac{\mathbf{v} \cdot \mathbf{D}_e\,\dot{\mathbf{w}}}{\mathbf{v} \cdot \mathbf{D}_e\,\mathbf{r} - y\,m\,||\mathbf{r}||}, \tag{26}$$

where $\mathbf{v} = \frac{\partial f}{\partial \boldsymbol{\sigma}}$, $y = \frac{\partial f}{\partial \sigma_{max}}$ and $m = \frac{\partial \sigma_{max}}{\partial u_p}$. Next, by substituting Eq. (26) into Eq. (23), it is possible to find the elasto-plastic constitutive matrix $\mathbf{D}_{ep}$ as

$$\mathbf{D}_{ep} = \mathbf{D}_e - \frac{\mathbf{D}_e\,\mathbf{r}\,\mathbf{v}^T\,\mathbf{D}_e}{\mathbf{v}^T\,\mathbf{D}_e\,\mathbf{r} - y\,m\,||\mathbf{r}||}. \tag{27}$$

The required derivatives for the calculation of $\mathbf{D}_{ep}$ are given by

$$y = -\tan\phi \tag{28}$$

$$\mathbf{v} = \left\langle \frac{\partial f}{\partial \sigma_n}, \frac{\partial f}{\partial \tau_s}, \frac{\partial f}{\partial \tau_t} \right\rangle = \left\langle \tan\phi, \frac{\tau_s}{\sqrt{\tau_s^2 + \tau_t^2}}, \frac{\tau_t}{\sqrt{\tau_s^2 + \tau_t^2}} \right\rangle \tag{29}$$

$$\mathbf{r} = \left\langle \frac{\partial g}{\partial \sigma_n}, \frac{\partial g}{\partial \tau_s}, \frac{\partial g}{\partial \tau_t} \right\rangle = \left\langle 2\sigma_n \tan^2\phi, 2\tau_s, 2\tau_t \right\rangle \quad \text{for} \quad \sigma_n > 0 \tag{30}$$

$$\mathbf{r} = \left\langle \frac{\partial g}{\partial \sigma_n}, \frac{\partial g}{\partial \tau_s}, \frac{\partial g}{\partial \tau_t} \right\rangle = \langle 0, 2\tau_s, 2\tau_t \rangle \quad \text{for} \quad \sigma_n \leq 0 \tag{31}$$

3.2 Stress update

The trial stress vector is found by using the elastic constitutive matrix, thus

$$\boldsymbol{\sigma}^{tr} = \boldsymbol{\sigma}^n + \mathbf{D}_e\,\Delta\mathbf{w}, \tag{32}$$

where the superscript n represents the current step. If $f(\boldsymbol{\sigma}^{tr}, \sigma_{max}^n) \leq 0$, the increment is elastic and the new

stress state is given by $\boldsymbol{\sigma}^{n+1} = \boldsymbol{\sigma}^{tr}$; otherwise, the increment will be elasto-plastic.

In unloading/reloading conditions, the elastic matrix $\mathbf{D}_e$ is considered to update the stresses. This is different from fracture damage models (e.g. Alfano and Crisfield 2001; Liu et al. 2012) where a secant stiffness is frequently used. Besides that, according to experimental data (Reinhardt and Cornelissen 1984; Chen et al. 2016; Stone and Webster Engineering Corporation 1992), specimens subjected to pure tension exhibit unloading curves with slopes varying between secant and almost initial stiffness. Also, unloading/reloading curves show considerable residual opening. For this reason, we consider that it is acceptable to use the elastic constitutive matrix under unloading/reloading conditions.

In turn, when the increment is elasto-plastic, the internal variable σ_{max} is modified and, the tip of the yield surface moves towards the origin. Also, the stress state, initially approximated by $\boldsymbol{\sigma}^{tr}$, will return to the yield surface following a direction perpendicular to the potential surface g, as shown in Fig. 8. The values of u_p, σ_{max} and $\boldsymbol{\sigma}$, as functions of the increment of relative displacements $\Delta\mathbf{w}$, are given by the following equations

$$u_p^{n+1} = u_p^n + \int_{t_n}^{t_{n+1}} \dot{u}_p \, dt = u_p^n + \Delta\lambda ||\mathbf{r}^{n+1}||, \tag{33}$$

$$\sigma_{max}^{n+1} = \sigma_{max}^n + \frac{\partial \sigma_{max}}{\partial u_p} \Delta\lambda ||\mathbf{r}^{n+1}||, \tag{34}$$

$$\boldsymbol{\sigma}^{n+1} = \boldsymbol{\sigma}^{tr} - \Delta\lambda \, \mathbf{D}_e \, \mathbf{r}^{n+1}. \tag{35}$$

Note that the identities $\Delta\mathbf{w}^{n+1} = \Delta\mathbf{w}_e^{n+1} + \Delta\mathbf{w}_p^{n+1}$ and $\Delta\mathbf{w}_p^{n+1} = \Delta\lambda \, \mathbf{r}^{n+1}$ were used in Eq. (32) in order to find Eq. (35).

According to the shape of the potential function g, we verify that $\text{sgn}\,(\sigma_n^{n+1}) = \text{sgn}\,(\sigma_n^{tr})$. This allows us to choose the right expression for $\mathbf{r}$ (see Eqs. 30 and 31). By substituting Eqs. (30) and (31) into Eq. (35) and isolating the stress components at step $n+1$ we get

$$\langle \sigma_n^{n+1}, \tau_s^{n+1}, \tau_t^{n+1} \rangle = \left\langle \frac{\sigma_n^{tr}}{1 + 2\Delta\lambda k_n \tan^2\phi}, \frac{\tau_s^{tr}}{1 + 2\Delta\lambda k_s}, \frac{\tau_t^{tr}}{1 + 2\Delta\lambda k_s} \right\rangle \quad \text{for} \quad \sigma_n^{tr} > 0, \tag{36}$$

and

$$\langle \sigma_n^{n+1}, \tau_s^{n+1}, \tau_t^{n+1} \rangle = \left\langle \sigma_n^{tr}, \frac{\tau_s^{tr}}{1 + 2\Delta\lambda k_s}, \frac{\tau_t^{tr}}{1 + 2\Delta\lambda k_s} \right\rangle \quad \text{for} \quad \sigma_n^{tr} < 0. \tag{37}$$

Equations (33) through (37) depend on $\Delta\lambda$, which in turn can be found using the consistency condition at step $n+1$,

$$f(\boldsymbol{\sigma}^{n+1}, \sigma_{max}^{n+1}) = \sqrt{(\tau_s^{n+1})^2 + (\tau_t^{n+1})^2} + (\sigma_n^{n+1} - \sigma_{max}^{n+1}) \tan\phi = 0. \tag{38}$$

This provides a nonlinear equation on $\Delta\lambda$ that can be solved using an iterative method. Applying the Newton-Raphson method we get

$$\Delta\lambda_{i+1} = \Delta\lambda_i - \frac{f(\boldsymbol{\sigma}^{n+1}, \sigma_{max}^{n+1})_i}{f'(\boldsymbol{\sigma}^{n+1}, \sigma_{max}^{n+1})_i}, \tag{39}$$

where i indicates the i-th iteration and $f'(\boldsymbol{\sigma}^{n+1}, \sigma_{max}^{n+1})$ is given by

$$\frac{\partial f^{n+1}}{\partial \Delta\lambda} = \frac{\partial f^{n+1}}{\partial \boldsymbol{\sigma}^{n+1}} \cdot \frac{\partial \boldsymbol{\sigma}^{n+1}}{\partial \Delta\lambda} + \frac{\partial f^{n+1}}{\partial \sigma_{max}^{n+1}} \frac{\partial \sigma_{max}^{n+1}}{\partial \Delta\lambda}. \tag{40}$$

We consider $\Delta\lambda_0 = 0$ as the initial guess for the iterative process. The corresponding derivatives are presented below, where the superscript $n+1$ was dropped for clarity:

$$\frac{\partial \boldsymbol{\sigma}}{\partial \Delta\lambda} = \left\langle \frac{-2k_n \tan^2\Phi \, \sigma_n^{tr}}{(1 + 2\Delta\lambda k_n \tan^2\Phi)^2}, \frac{-2k_s \tau_s^{tr}}{(1 + 2\Delta\lambda k_s)^2}, \frac{-2k_s \tau_t^{tr}}{(1 + 2\Delta\lambda k_s)^2} \right\rangle \quad \text{for } \sigma_n^{tr} > 0, \tag{41}$$

$$\frac{\partial \boldsymbol{\sigma}}{\partial \Delta\lambda} = \left\langle 0, \frac{-2k_s \tau_s^{tr}}{(1 + 2\Delta\lambda k_s)^2}, \frac{-2k_s \tau_t^{tr}}{(1 + 2\Delta\lambda k_s)^2} \right\rangle \quad \text{for } \sigma_n^{tr} \le 0, \tag{42}$$

$$\frac{\partial \mathbf{r}}{\partial \Delta\lambda} = \left\langle \frac{-4k_n \tan^4\Phi \, \sigma_n^{tr}}{(1 + 2\Delta\lambda k_n \tan^2\Phi)^2}, \frac{-4k_s \tau_s^{tr}}{(1 + 2\Delta\lambda k_s)^2}, \frac{-4k_s \tau_t^{tr}}{(1 + 2\Delta\lambda k_s)^2} \right\rangle \quad \text{for } \sigma_n^{tr} > 0, \tag{43}$$

$$\frac{\partial \mathbf{r}}{\partial \Delta\lambda} = \left\langle 0, \frac{-4k_s \tau_s^{tr}}{(1 + 2\Delta\lambda k_s)^2}, \frac{-4k_s \tau_t^{tr}}{(1 + 2\Delta\lambda k_s)^2} \right\rangle \quad \text{for } \sigma_n^{tr} \le 0, \tag{44}$$

$$\frac{\partial \sigma_{max}}{\partial \Delta\lambda} = \frac{\sigma_{max}}{u_p} \left(||\mathbf{r}|| + \Delta\lambda \mathbf{r} \cdot \frac{\partial \mathbf{r}}{\partial \Delta\lambda} \right) \tag{45}$$

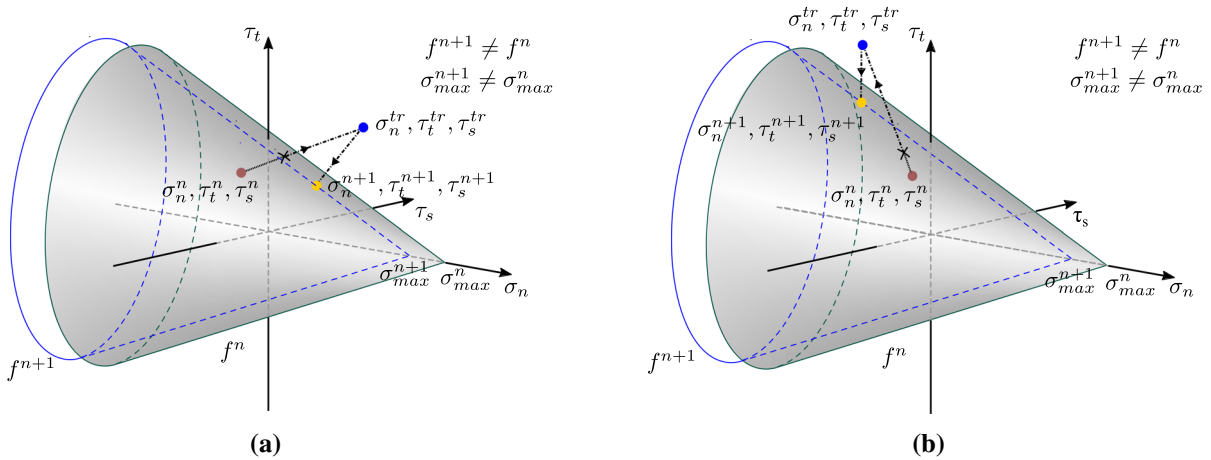


Fig. 8 Stress update in the elasto-plastic regime: **a** return to the yield surface when $\sigma_n^{tr} > 0$; **b** return to the yield surface when $\sigma_n^{tr} \leq 0$

3.3 Plastic update for $u_p \geq u_c$

When the crack achieves the critical opening ($u_p \geq u_c$), there is no more tensile strength, thus $\sigma_{max} = 0$. Then, if at step $n+1$ the crack gets open ($w^{n+1} \geq u_c$) the trial stress should return to the origin (Fig. 9). However, at this stage, the potential function is reduced to a line and the derivatives at the origin are undefined. To handle this situation, an expression for $||\mathbf{r}^{n+1}||\Delta\lambda$ is required in order to finalize the stress update process applying Eqs. (33) and (34). Thus, in this case, expanding Eq. (35) we obtain

$$\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} \sigma_n^{tr} \\ \tau_s^{tr} \\ \tau_t^{tr} \end{pmatrix} - \Delta\lambda \begin{bmatrix} k_n & 0 & 0 \\ 0 & k_t & 0 \\ 0 & 0 & k_t \end{bmatrix} \begin{pmatrix} r_n^{n+1} \\ r_s^{n+1} \\ r_t^{n+1} \end{pmatrix}, \tag{46}$$

which leads to

$$||\mathbf{r}^{n+1}||\Delta\lambda = \left\| \left\langle \frac{\sigma_n^{tr}}{k_n}, \frac{\tau_s^{tr}}{k_t}, \frac{\tau_t^{tr}}{k_t} \right\rangle \right\|. \tag{47}$$

Besides, and not less important, it is worth noting that while the crack is open, it should be considered $\mathbf{D}_{ep} = \mathbf{0}$ at the stage of computing the element stiffness matrix.

4 Solution procedure

The solution of the system of nonlinear equations can be performed using a conventional incremental solver. Nevertheless, this work's solution procedure includes some techniques to improve convergence, reduce the number of increments, and correct the results drift from the equilibrium path. They are described in the following subsections.

4.1 Application of Runge–Kutta methods

A simple solution procedure for a finite element system of nonlinear equations will use, for example, an incremental scheme combined with the Newton–Raphson method, approximating the stiffness matrix at the beginning of each loading step. This approach applied with displacement control can solve nonlinear systems that present softening behavior. However, it may fail in strongly nonlinear problems due to high truncation errors; for example, when the analysis approaches the peak load, where the material stiffness changes drastically.

On the other hand, the application of Runge–Kutta methods to the solution of systems of equations aims to improve the accuracy based on two or more evaluations of the stiffness matrix. For example, using the modified Euler method, two evaluations for the stiffness matrix are required. They can be written as $\mathbf{K}_1 = \mathbf{K}(\mathbf{U}^n)$ and $\mathbf{K}_2 = \mathbf{K}(\mathbf{U}^n + \Delta\mathbf{U})$, where $\mathbf{U}^n$ is the global displacements vector at the beginning of the step and $\Delta\mathbf{U}$ is the displacements increment vector. In this case, the final approximation of the stiffness matrix is given by

$$\bar{\mathbf{K}} = \frac{1}{2}(\mathbf{K}_1 + \mathbf{K}_2). \tag{48}$$

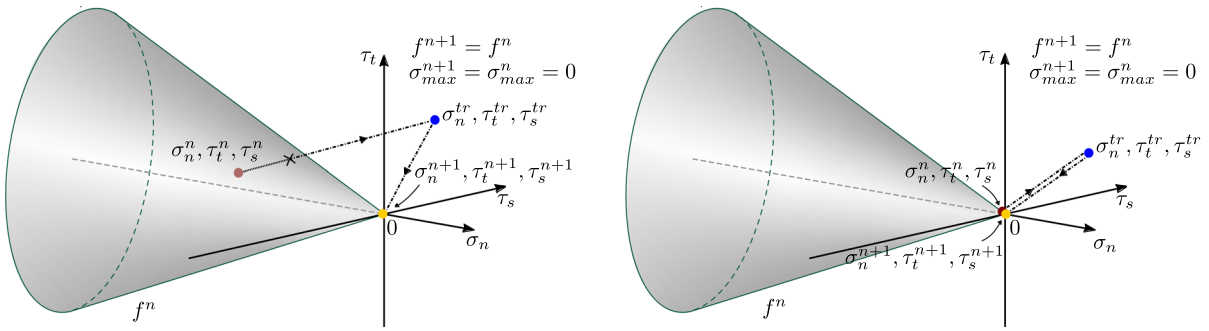


Fig. 9 Stress update when the crack is open at step $n+1$: **a** the crack faces are still in contact at step n; **b** the crack faces are already separated at step n

Another method that involves the same amount of calculations as the previous one but provides greater accuracy is the Ralston's method (Ralston 1962). In this case, the two stiffness matrices are $\mathbf{K}_1 = \mathbf{K}_1(\mathbf{U}^n)$ and $\mathbf{K}_2 = \mathbf{K}_1(\mathbf{U}^n + \frac{2}{3}\Delta\mathbf{U})$, and the final stiffness matrix is given by

$$\bar{\mathbf{K}} = \frac{1}{4}\mathbf{K}_1 + \frac{3}{4}\mathbf{K}_2. \tag{49}$$

In this paper, the Ralston method (Eq. 49) was employed to improve the approximation of the global stiffness matrix which is later used to find the unknown displacements. The use of this method allowed us to obtain improved accuracy which was reflected in a fewer number of increments.

4.2 Automatic stepping

In a nonlinear incremental analysis, the load and displacement increments at a given step are usually given by

$$\Delta\mathbf{F} = \Delta T\, \mathbf{F}_{ext} \quad \text{and} \tag{50}$$

$$\Delta\mathbf{U} = \Delta T\, \mathbf{U}. \tag{51}$$

where $T \in [0, 1]$ is the pseudo-time for the current analysis stage and ΔT is the step size used to scale the external forces $\mathbf{F}_{ext}$ and displacements $\mathbf{U}$ vectors. When constant step size is employed, usually, very small ΔT values are required to get the truncation error (err) from all increments under the specified tolerance (tol) since the analysis will stop at the first case where $err > tol$. On the other hand, an automatic stepping algorithm modifies ΔT such that it is increased after a successful increment ($err \leq tol$) and decreased when an increment trial fails. In this scheme, the increment size at step $n+1$ can be calculated based on the previous increment as

$$\Delta T^{n+1} = q\, \Delta T^n, \tag{52}$$

where q is a factor that aims to limit the truncation error defined here as $err = ||\Delta\mathbf{F}_{ext} - \Delta\mathbf{F}_{int}||$. The basic idea is to use $q \geq 1$ when the last increment trial was successful and $q < 1$ otherwise. The expression for q usually takes the form $c\sqrt{\frac{tol}{err}}$, where c is an adjustable coefficient (see e.g. Abbo and Sloan 1996; Ho et al. 1997; Sheng et al. 2002). In this research we propose the expression

$$q = \left(1 + \tanh\left(\log_{10}\frac{tol}{err}\right)\right). \tag{53}$$

The equation above is quite arbitrary but has the following properties. First, it is a continuous function with values in the interval (0, 2). It yields 1 when $err = tol$ and values smaller than 1 when $err > tol$. Also, it tends to 2 when err tends to 0. Since, this expression varies smoothly, it can be used to gradually adjust the increment size as follows. When the last increment trial converged successfully ($err \leq tol$ and $q \geq 1$), the increment size grows following Eq. (52), otherwise ($err > tol$ and $q < 1$), the increment size decreases according to

$$\Delta T_{n+1} = \max(0.2,\ \min(0.9,\ q))\ \Delta T_n. \tag{54}$$

In this case, the scaling factor is limited to 0.2 to avoid tiny increments, and to 0.9 to favor convergence when $err \approx tol$.

4.3 Drift correction

Solutions of nonlinear equations, especially those obtained from incremental schemes, tend to drift from the equilibrium path as residual forces (unbalanced forces between the externally applied forces and the forces supported by the internal stresses) are accumulated along with the increments. To reduce drifting, one strategy is to add the residual forces from one increment to the next. However, as pointed out by Abbo and Sloan (1996), this procedure is not efficient when combined with an automatic stepping scheme since the residual forces contribution is independent of the step size ΔT. Also, for strongly nonlinear behavior, the new increment error frequently exceeds the specified tolerance due to the residual forces effect. To deal with this problem, the strategy used in this paper is to spread cumulated residues over a fraction (ΔT_{cr}) of the analysis time, which may involve several upcoming increments. This fraction can be defined as

$$\Delta T_{cr} = \min(\Delta T_{rspan}, 1 - T), \tag{55}$$

where ΔT_{rspan} is the maximum fraction of the analysis to apply the residual forces obtained from one increment (e.g. 0.01 or 1%).

Assuming that $\mathbf{R}_c^n$ is a vector containing the cumulated residues up to the end of step n, at step $n+1$, the vector with the increments of external forces is modified according to

$$\Delta \mathbf{F}_{ext}^{n+1} = \Delta T^{n+1} \mathbf{F}_{ext} + \gamma \mathbf{R}_c^n, \tag{56}$$

where $\gamma = \min(1, \Delta T^{n+1}/\Delta T_{cr})$ is the coefficient used to obtain a fraction of $\mathbf{R}_c^n$ to be applied at increment $n+1$. If step $n+1$ is successful, then the cumulated residues vector is updated considering the residue from step $n+1$ and the remaining fraction, not yet applied, from $\mathbf{R}_c^n$. Thus,

$$\mathbf{R}_c^{n+1} = \mathbf{R}^{n+1} + (1 - \gamma)\, \mathbf{R}_c^n. \tag{57}$$

In the case of failure, the cumulated residual vector is not modified and the increment is rejected. Finally, in both cases, a new step size for the next increment is calculated according to the automatic stepping algorithm.

Besides providing results with a better agreement with experimental data, the drift correction procedure presented here reduces the chance of increments failure caused by the reapplication of residual forces.

5 Numerical simulations

Prior to the simulation of application examples, the proposed constitutive model was verified using patch tests (extension and shear) and, in the sequence, a mesh refinement test is presented to study mesh dependency. Later, the model was applied to the fracture simulation of concrete specimens. The results were compared with the experimental data available. All the examples present two laboratory experimental curves. These curves represent repeated experiments using the same material and geometry.

During the simulations, the mechanical behavior of bulk elements was considered as linear elastic. In turn, all cohesive elements were simulated using the proposed model considering the Hordijk's traction–separation curve. The simulations were performed using the Amaru library (https://github.com/NumSoftware/Amaru) which includes the implementation of the proposed model.

5.1 Mesh generation

All meshes were elaborated with the aid of Gmsh (Geuzaine and Remacle 2009), a 2D and 3D finite element mesh generator. After the generation stage, cohesive elements were introduced using the algorithm explained in Sect. 2.2. Although it is possible to limit the location of cohesive elements, we decided to place them over the whole mesh as a way to verify that cracks will only happen where expected. The meshes from all examples were refined enough to provide good, or fairly good, agreement with the experimental data.

It is worth mentioning that the mesh elaboration process can take into consideration the minimum number of elements within the FPZ (see e.g. Davila et al. 2001; Falk et al. 2001; Turon et al. 2007; Harper and Hallett 2008). Considering the material properties from the simulated examples, the estimates for the cohesive length, as proposed by Hillerborg et al. (1976), provided values greater than 27 cm. In all cases, this length was enough to contain more than three elements, as suggested by Harper and Hallett (2008). In this regard, we

consider a good practice to perform local refinements in regions where cracks are expected.

Regarding the mesh elements, the type of bulk elements used were 4-node tetrahedrons with 4 integration points, 5-node pyramids with 5 integration points, 6-node wedges with 9 integration points and 8-node hexahedrons with 8 integration points. As for the cohesive elements, we used 6-node joints with 3 integration points and 8-node joints with 4 integration points. It is worth mentioning that some trials were performed using 6-node interface elements (triangular shape) with one integration point. However, inconsistent displacements were obtained, similar to the spurious deformations observed in meshes with elements with reduced integration. Although triangular elements require only one integration point in two-dimensional stress analyses, this is not enough for 6-node interface elements due to the higher degree of the integrand used to compute the stiffness matrix.

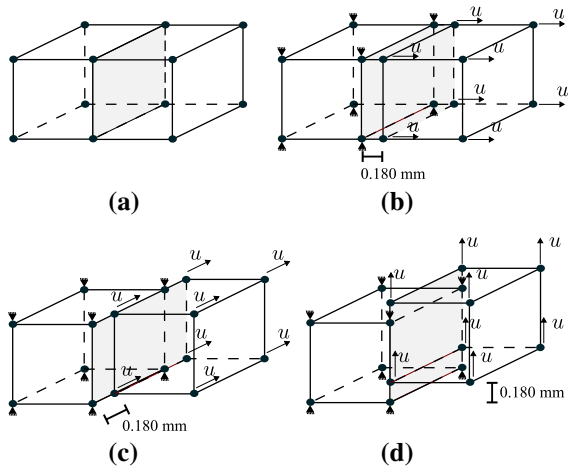


Fig. 10 Patch tests: **a** initial configuration; **b** extension test; **c** first shear test; **d** second shear test

5.2 Material parameters

For the numerical analysis, the material parameters for the presented model can be related to bulk and the cohesive elements. Since bulk elements are considered linear elastic, only Young's modulus E and Poisson's ratio ν are required. For the cohesive elements, in addition to the elastic parameters, the following are required: the tangent of the internal friction angle $\tan(\phi)$, the tensile strength σ^0_{max}, and the fracture energy G_F. In the numerical examples, except from $\tan(\phi)$ and α, all other parameters were taken from the corresponding experiment publication or publications that perform the numerical simulations.

Regarding the internal friction angle for concrete, experimental results (see e.g. Fujita et al. 1997; Stone and Webster Engineering Corporation 1992; Jebli et al. 2018) show values ranging from 54° to 57°. These angles provide $\tan\phi$ between 1.37 and 1.53, approximately. In this sense, and with no additional information, we found reasonable to adopt $\tan(\phi) = 1.4$ for all application examples. Regarding this value, no negative effects in the results were observed during the simulations. In turn, the α parameter, as stated in Sect. 3, aims to provide negligible elastic displacements prior to the crack formation. Too small values provide excessive elastic displacements, while too high values provide high stiffness coefficients that lead to ill-conditioned systems of equations. During the analyses, we observed that $\alpha = 5$ effectively provides negligible elastic displacements while avoids numerical problems.

5.3 Patch tests

To verify the behavior of the proposed model, patch tests in extension and shear are presented. In these analyses, only two bulk elements linked by one cohesive element are considered. The bulk elements are given by 8-noded regular hexahedrons with 10 mm length. For each case, one bulk element is kept restrained while the other one is submitted to rigid body motion; thus, restraining all deformations to the cohesive element to simulate extension and shear. Figure 10 shows the initial configuration and different displacement conditions that have been studied. The material parameters used in these analyses are presented in Table 1.

For the extension analysis, an imposed displacement $u = 0.180$ mm was applied (Fig. 10b). The relative displacements and corresponding stresses were logged at one integration point from the cohesive element. Figure 11 shows the resulting curves $w - \sigma_n$ and $u_p - \sigma_n$. As seen, the interface starts showing a highly stiff behavior, and once the tensile strength is reached, it shows a softening behavior comparable to Hordijk's model.

As for the shear analysis, two tests were performed, as shown in Figs. 10c and d. In both cases, a displacement ($u = 0.180$ mm) was applied in a direction parallel to the interface. As expected, the results from both

Table 1 Patch tests: parameters of the material and interface elements

Softening curve	E (GPa)	ν	$\tan(\phi)$	α	σ^0_{max} (MPa)	G_F (N/m)	u_c (mm)
Hordijk	27.0	0.20	1.4	5	2.40	80.0	0.1712

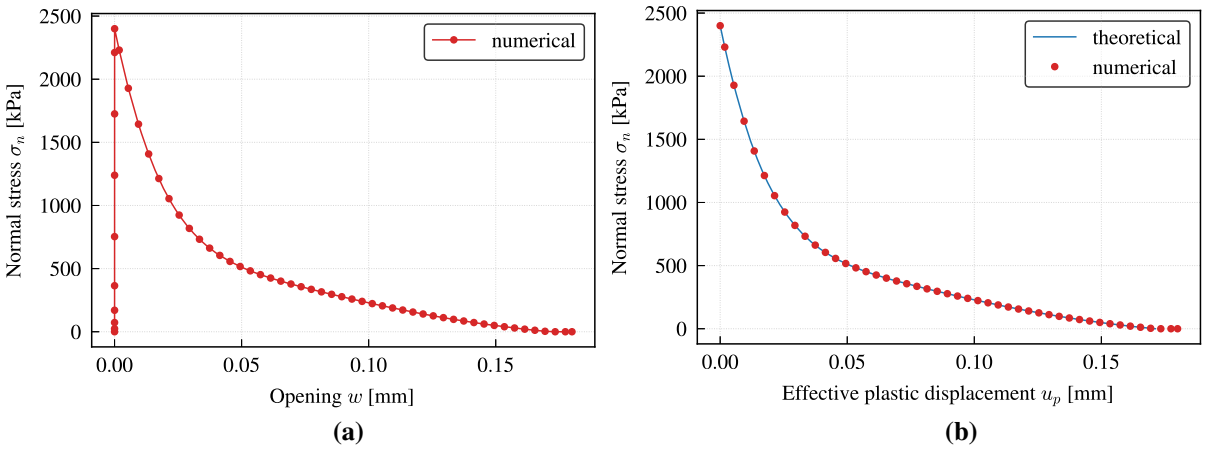


Fig. 11 Extension test: **a** crack opening versus normal stress; **b** effective plastic displacement versus normal stress

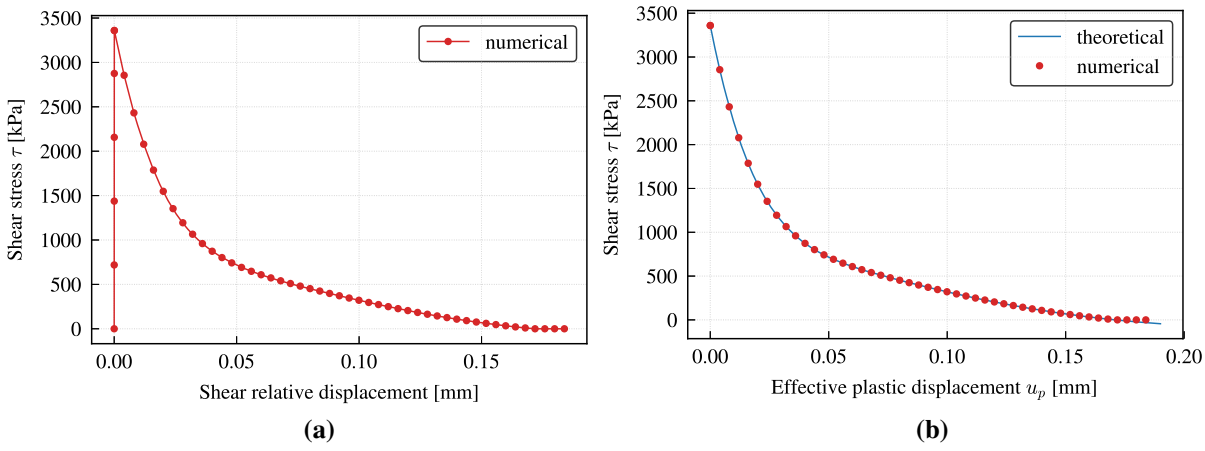


Fig. 12 Shear test: **a** displacement parallel to the interface plane versus shear stress; **b** effective plastic displacement versus shear stress

shear tests were similar. Figure 12 presents the results of shear stress (τ_s) versus shear displacement and effective plastic displacement. Again, the numerical values are in good agreement with the theoretical values.

5.4 Mesh refinement test

Finite element simulations using the discrete approach may suffer from mesh sensitivity, where different mesh discretizations provide different results. To avoid this issue, the stiffness from a cohesive element depends on the surrounding bulk element sizes (representative length). In practice, this means that smaller cohesive elements provide slightly stiffer behavior than larger ones. To better evaluate this approach to counter mesh dependency, we perform a mesh refinement test. Figure 13 shows the domain given by a notched block subjected to tensile cracking. Five tetrahedral non-

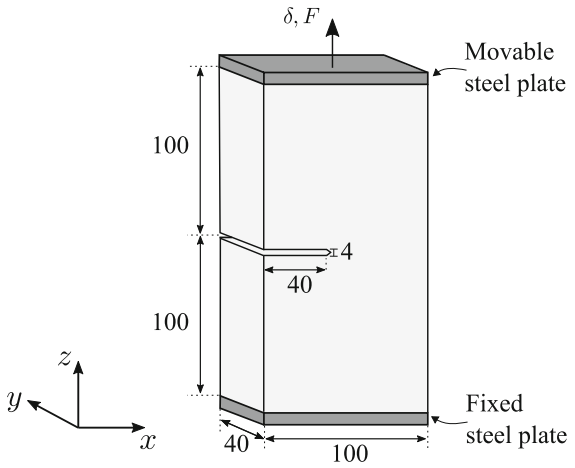


Fig. 13 Refinement test: boundary conditions and material properties. All measures are in millimeters

Table 2 Refinement test: summary of mesh attributes

Mesh	Bulk elements	Cohesive elements	Nodes	DOFs	h_{med}
Mesh 1	659	1112	2636	7908	0.0079
Mesh 2	1685	2977	6740	20,220	0.0059
Mesh 3	2556	4548	10,224	30,672	0.0051
Mesh 4	5337	9696	21,348	64,044	0.0041
Mesh 5	9234	17,078	36,936	110,808	0.0031

Table 3 Refinement test: material parameter for bulk and cohesive elements

E (GPa)	ν	$\tan(\phi)$	α	σ^0_{max} (MPa)	G_F (N/m)
32.0	0.20	1.4	5	3.30	60.0

structured meshes comprising different element sizes were used, as shown in Fig. 14. In turn, Table 2 presents a summary of the mesh attributes. Due to geometry constraints and the refinement near the notch, the element size varies within each mesh. For this reason, the median value h_{med} is shown in Table 2. For example, note that in the first mesh (coarse), h_{med} is more than twice compared to the fifth mesh (most refined). We can also note a huge difference in degrees of freedom. As for the material parameters used in the simulations, they are shown in Table 3.

During the analysis, the base plate is kept fixed, while prescribed vertical displacements are applied at the top plate. The resultant force at the top plate as well as the crack mouth opening displacement (CMOD) were logged. Although the meshes comprise different numbers of cohesive elements, it is expected to get similar load–CMOD curves before and after the peak load, as shown in Fig. 15. This can be explained by the following reasons. Before the peak, the deformability is mainly controlled by bulk elements while cohesive elements present negligible deformability. On the other hand, after the peak, the deformability is mainly controlled by the cracks. Note that before the peak, all curves follow pretty much the same path. This demonstrates that the proposed formulation practically eliminates mesh dependency on the elastic regime. Regarding the peak load, meshes 2 to 5 provided sim-

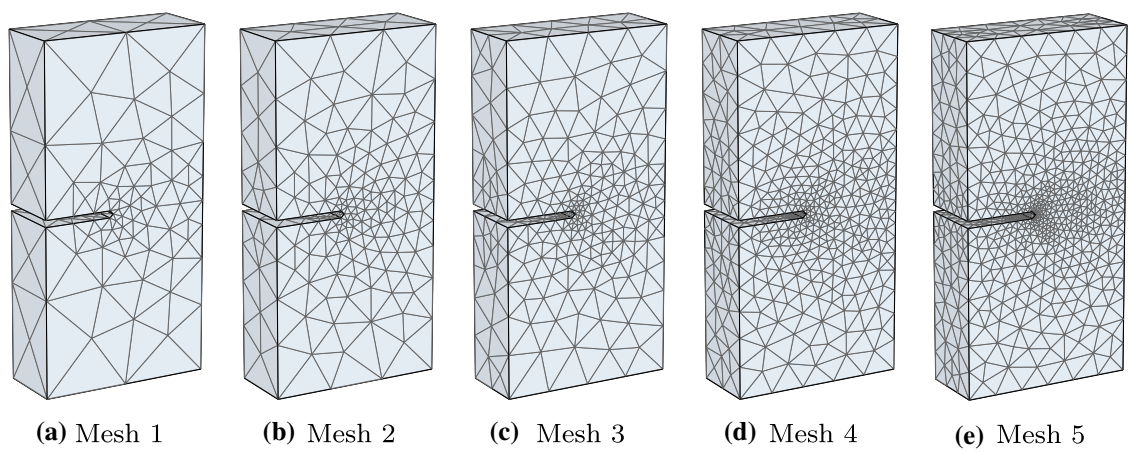


Fig. 14 Refinement test: finite element meshes

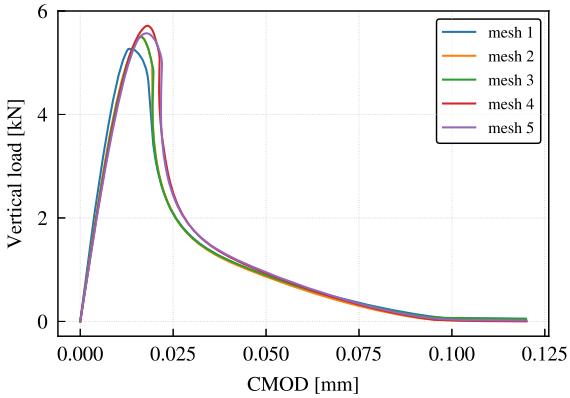


Fig. 15 Refinement test: load-CMOD curves

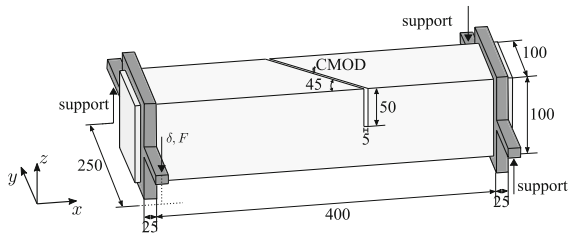


Fig. 17 Example 1: test setup. All measures are in millimeters

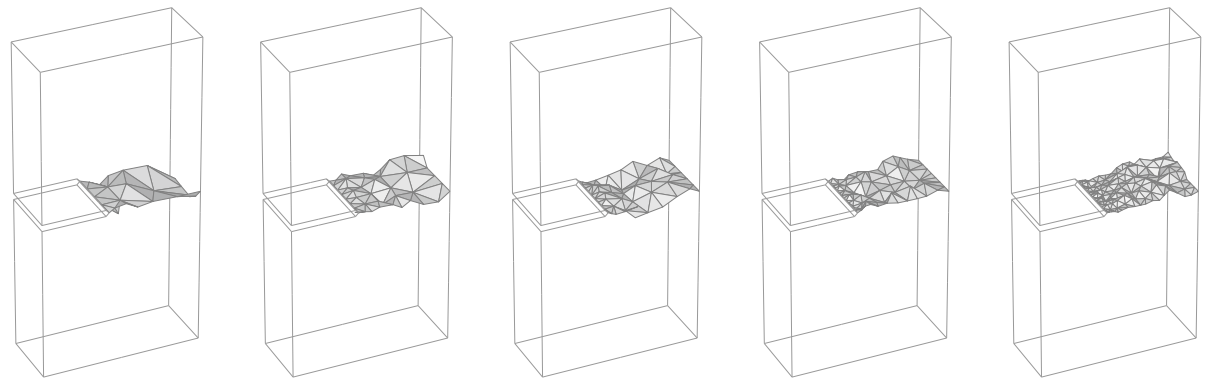
Table 4 Example 1: material parameter for bulk and cohesive elements

E (GPa)	ν	$\tan(\phi)$	α	σ^0_{max} (MPa)	G_F (N/m)
35.0	0.20	1.4	5	2.30	80.0

ilar values, around 5.6 kN. This is not the case for the first mesh (coarse), which presented a smaller peak value. After the peak load, all curves are close and show the same nonlinear pattern, with no waves nor sudden changes. We can also observe that the most refined meshes, 4 and 5, present similar peak values and almost identical post-peak curves. These results show that the proposed method tends to reduce mesh dependency. A specific study on the limit of element sizes was not performed. Thus, for practical numerical analyses, as in other finite element applications, a refinement test is suggested in order to obtain a functional mesh. A view of the crack surfaces, Fig. 16, shows that they tend to evolve towards a flat geometry. This behavior was expected since the element sizes are decreasing and the domain is subjected to symmetrical conditions.

5.5 Example 1: beam with a notch subjected to torsion

This example simulates the experimental test presented by Brokenshire (1995). The typical test setup is shown in Fig. 17 where a notched beam is clamped by two metallic frames. The beam is supported at the frame extremities, except at one corner where a prescribed displacement is applied to induce torsion in the sample. Due to the lack of symmetry, this experiment cannot be simulated by a two-dimensional simplification; thus, three-dimensional modeling is mandatory. The material parameters used in the simulation were extracted from the work of Jefferson et al. (2004) and are presented in Table 4.



(a) Mesh 1 **(b)** Mesh 2 **(c)** Mesh 3 **(d)** Mesh 4 **(e)** Mesh 5

Fig. 16 Refinement test: final crack for each mesh

Two numerical simulations were performed using different element types. In order to introduce randomness in the shape of the crack path, unstructured meshes were used. Table 5 presents a summary of the finite element meshes. The first mesh uses tetrahedral elements, while the second one uses a combination of tetrahedral and pyramidal elements.

During the finite element analysis, the reaction force due to the imposed displacement and CMOD at the notch were logged for further analysis. Figure 18 shows the numerical and experimental curves of load versus CMOD. The overall behavior of the numerical curves is in fair agreement with the experimental data. Both numerical curves underestimate the peak load but present a good representation of the softening branch, especially when compared with the first experimental curve. Few differences can be observed between the use of tetrahedrons and the combination of tetrahedrons and pyramids. Although, in this case, the latter permitted using fewer elements. Table 6 shows a summary of peak loads (F_{max}) and corresponding CMOD values (δ) obtained from the numerical results and the experimental curves presented by Brokenshire. Taking the first experiment results as reference, the relative differences are computed and included in the table. Note that the discrepancies between experiments and between the first experiment and numerical results are all less than one order of magnitude. Therefore, we can consider that the obtained numerical results are within an acceptable range. Furthermore, Fig. 18 also shows numerical curves from Jefferson et al. (2004), Gasser and Holzapfel (2006) and Ferté et al. (2016), which were obtained using the plastic-damage model, PUFEM (Partition Unity Finite Element Method) and XFEM, respectively. Comparing with the experimental curves, we can see that results from Jefferson et al. show a good prevision of the peak load and a brittle behavior after that; however, the data is only available up to $CMOD = 0.4$ mm. On the other hand, Gasser et al. presented two curves for this test obtained using two

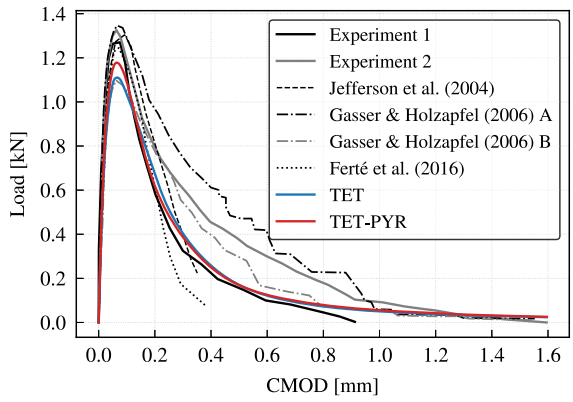


Fig. 18 Example 1: experimental and numerical results

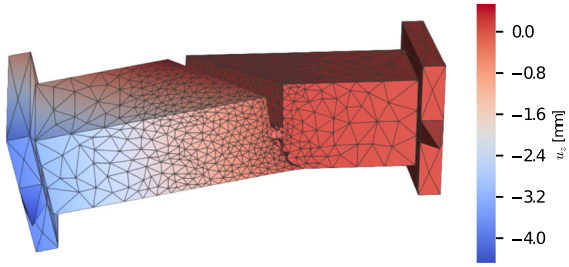


Fig. 19 Example 1: deformed mesh displaying the u_z field for the TET analysis

different sets of material parameters. As can be seen, one curve from Gasser et al. overestimates the softening branch while the other underestimates the peak load. Finally, the results from Ferté et al. present a peak load compatible with the experiments, but only part of the softening branch is available.

Figure 19 shows the deformed mesh at the end of the analysis that uses tetrahedral elements. The mesh displays the vertical displacement field and the deformation was amplified using a scale factor of 15. Furthermore, Fig. 20 presents the crack surface evolution for different prescribed displacement levels considering the cohesive elements where the average value of w_n was greater than $0.5\,u_c$.

Table 5 Example 1: summary of mesh attributes

Model	Element types	Bulk elements	Cohesive elements	Nodes	DOFs
TET	Tetrahedral	8706	15,964	34,824	104,472
TET-PYR	Tetrahedral + pyramidal	7744	15,349	31,585	94,755

Table 6 Example 1: comparison of numerical and experimental results

Experiment/model	Increments	F_{max} (kN)	CMOD δ (mm)	$\frac{\lvert\Delta F_{max}\rvert}{F_{max}}$ %	$\frac{\lvert\Delta\delta\rvert}{\delta}$ %
Experiment 1	–	1.27	0.0759	–	–
Experiment 2	–	1.33	0.0627	4.4	17.4
TET	504	1.11	0.0663	12.6	12.7
TET-PYR	710	1.18	0.0647	7.2	14.7

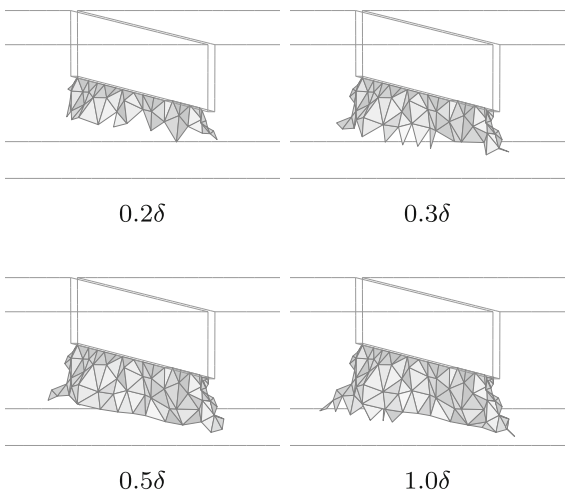


Fig. 20 Example 1: crack evolution according the prescribed displacement for the TET analysis

5.6 Example 2: circular section beam with a notch subjected to torsion

This second application example simulates a notched beam with a circular section tested by Brokenshire (1995). Two circular frames clamp the beam, and the whole setup is supported at three points, as shown in Fig. 21. In a similar fashion to the square section beam, a prescribed displacement is applied at one end of the leftmost frame to induce torsion to the sample. The material parameters used in the simulation are the same as in Example 1, given that both experiments were performed by the same author using the same material. Two tetrahedral meshes with different refinement levels are employed. Table 7 presents a summary of mesh features, including the median value of the element size h_{med}.

Along with the simulation, the reaction force due to the imposed displacement and the CMOD at the notch

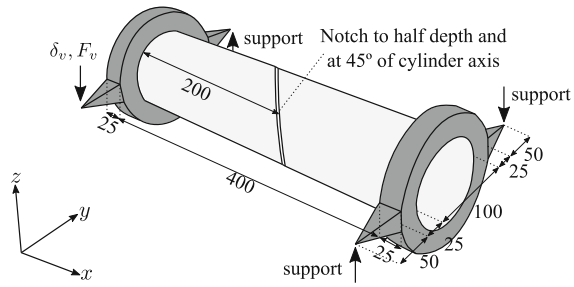


Fig. 21 Example 2: test setup. All measures are in millimeters

Table 7 Example 2: summary of mesh attributes

Model	Bulk elements	Cohesive elements	Nodes	DOFs	h_{med} (m)
TET1	1795	3104	7180	21,540	0.0130
TET2	4398	8578	17,926	53,778	0.0088

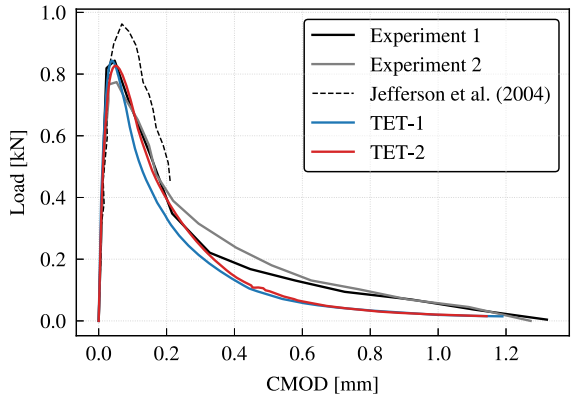


Fig. 22 Example 2: experimental and numerical results

were logged. Figure 22 shows the numerical and experimental curves of load versus CMOD. We can observe good agreement between numerical and experimental curves from the results, especially when compared with the first experiment. Also, both numerical curves are very similar, even though the second analysis has

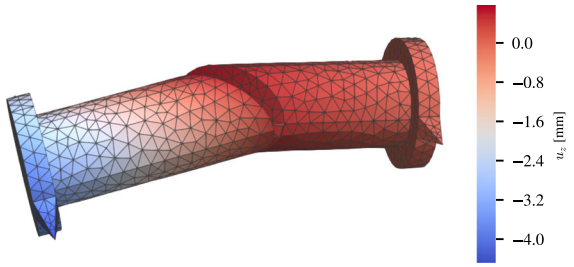


Fig. 23 Example 2: deformed mesh displaying the u_z field for the TET2 analysis

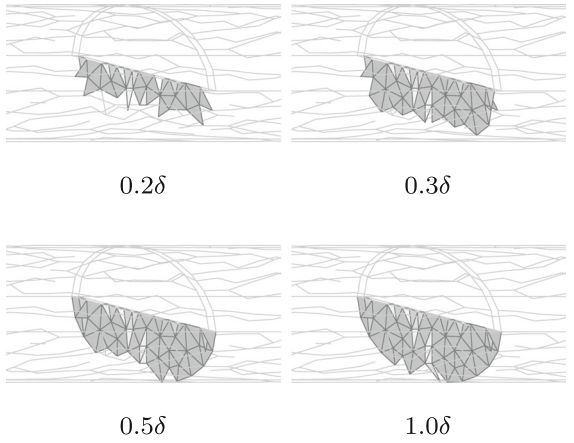


Fig. 24 Example 2: crack surface evolution according to the prescribed displacement for the TET2 analysis

more than double the elements. This suggests that both curves are close to a convergent curve.

Table 8 shows a comparative analysis of the peak load (F_{max}) and the corresponding CMOD (δ) against the first experimental curve. Numerical peak loads show excellent agreement. In turn, corresponding CMOD values show differences between 10 and 15%, approximately. Nonetheless, note that the discrepancies between experimental curves are also in a similar range. For the sake of comparison with other authors, Fig. 22 also shows the numerical results from Jefferson et al. (2004) that used a plastic-damage-contact model. Note that the peak and corresponding CMOD values were overestimated. Also, only part of the softening curve was available.

Figure 23 shows the deformed mesh at the end of the TET2 analysis using a magnification factor of 15. The figure also shows the u_z field for better visualization. As a means to observe the crack evolution in the TET2 analyses, Fig. 24 shows the cohesive elements where the average value of w_n was greater than $0.5\,u_c$.

5.7 Example 3: beam with an eccentric notch and loading

The third example analyzes the experimental test of a notched beam investigated by Feist and Hofstetter (2007). The corresponding setup is shown in Fig. 25, where an eccentric displacement is imposed at the beam bottom with the aid of a steel plate. The sample is mainly subjected to mode I fracture; however, note that this example is not symmetrical and cannot be simplified into a two-dimensional representation. The required material parameters were extracted from the work of Gasser (2007) and are presented in Table 9.

For the finite element analysis, two unstructured meshes were used. Table 10 presents a summary of the corresponding element meshes. During the numerical analysis, the reaction force at the steel plate location and

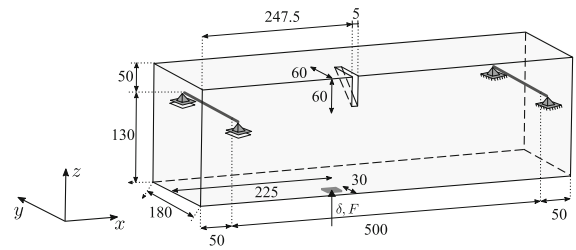


Fig. 25 Example 3: test setup. All measures are in millimeters

Table 8 Example 2: comparison of numerical results and experimental data

Experiment/model	Increments	P_{max} (kN)	CMOD δ (mm)	$\frac{\lvert\Delta P_{max}\rvert}{P_{max}}$ %	$\frac{\lvert\Delta\delta\rvert}{\delta}$ %
Experiment 1	–	0.844	0.0470	–	–
Experiment 2	–	0.774	0.0528	8.25	12.4
TET1	530	0.843	0.0398	0.04	15.3
TET2	196	0.827	0.0521	1.94	10.9

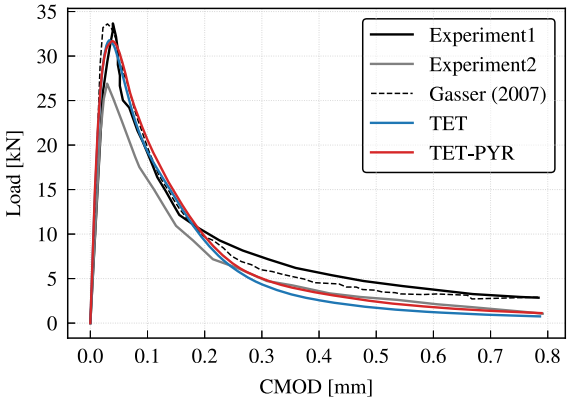


Fig. 26 Example 3: experimental and numerical results

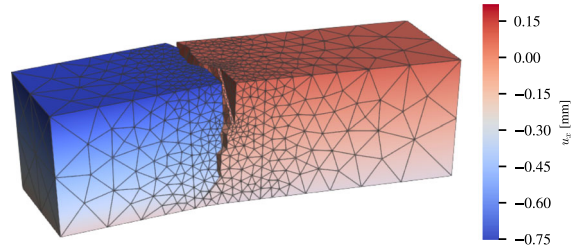


Fig. 27 Example 3: deformed mesh of the TET analysis displaying the horizontal displacement field

Table 9 Example 3: material parameter for bulk and cohesive elements

E (GPa)	ν	$\tan(\phi)$	α	σ^0_{max} (MPa)	G_F [N/m]
37.3	0.19	1.4	5	3.05	106

the CMOD at the notch were logged. The load versus CMOD curves from the numerical and experimental results are presented in Fig. 26. As can be seen, both numerical analyses present quite similar results and good agreement compared to the experiments. In this case, the introduction of pyramid elements allows fewer elements and consequently, fewer degrees of freedom without compromising the results. Table 11 shows a comparison of the peak load (F_{max}) and corresponding CMOD (δ), taking the first experiment as reference. Good agreement can be observed for the peak load values. More significant differences will be observed if the numerical results are compared with the second experiment curve. Nonetheless, it is worth to mention that there is a considerable difference between experiment curves in terms of peak loads and corresponding CMOD values. Figure 26 also includes the numerical curve obtained by Gasser (2007), who used the PUFEM. This curve also shows good agreement with the experiments data; however, we consider that the model presented here is more straightforward in terms of formulation since it is based on conventional finite elements and conventional plasticity.

Figure 27 shows the deformed mesh of the TET analyses at 90% of applied displacements. For better crack visualization, the figure shows the u_x field, and the deformation is amplified using a factor of 20. In turn, Fig. 28 allows seeing the crack evolution by displaying all cohesive elements where $w_n > 0.5\, u_c$.

Table 10 Example 3: summary of mesh attributes

Model	Element types	Bulk elements	Cohesive elements	Nodes	DOFs
TET	Tetrahedral	5766	10,621	23,026	69,078
TET-PYR	Tetrahedral + Pyramidal	3871	7705	15,798	47,394

Table 11 Example 3: comparison of numerical results and experimental data

Experiment/model	Increments	F_{max} [kN]	CMOD δ [mm]	$\frac{\lvert\Delta F_{max}\rvert}{F_{max}}$ %	$\frac{\lvert\Delta\delta\rvert}{\delta}$ %
Experiment 1	–	33.67	0.0395	–	–
Experiment 2	–	26.90	0.0293	20.1	25.6
TET	1037	31.78	0.0338	5.6	14.5
TET-PYR	1046	31.68	0.0389	5.9	1.5

Table 12 Example 4: material parameter for bulk and cohesive elements

E (GPa)	ν	$\tan(\phi)$	α	σ^0_{max} (MPa)	G_F (N/m)
30.0	0.20	1.4	5	3.30	100

5.8 Example 4: double notched square plate

This example analyzes the experimental test studied by Nooru-Mohamed (1992). The experiment's purpose was to study the cracking properties of concrete in mixed mode, so it induces slanted cracks that start at notches and go towards the specimen's center. The test setup is shown in Fig. 29. Note that the right side below the notch and the bottom are bonded to a fixed steel plate. In turn, the left side above the notch and the top are bonded to a movable steel plate. During the experiment, horizontal δ_h and vertical δ_v controlled displacements (with equal rates) were simultaneously applied at the movable steel plate. The figure also shows the positions where displacements were logged with the aid of LVDTs. The required material parameters were extracted from the original work of Nooru-Mohamed and are presented in Table 12.

In the numerical analyses, two unstructured meshes were used using wedge and hexahedral elements. Table 13 presents a summary containing the mesh attributes. Note that the minimum r_{min} and maximum r_{max} values for the elements aspect ratio are included. In this work, the aspect ratio is defined as the length of the largest edge divided by the shortest one.

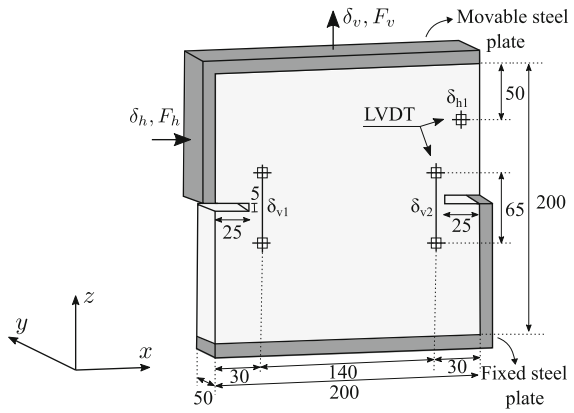


Fig. 29 Example 4: test setup. All measures are in millimeters

During the finite element analysis, the reaction forces F_h and F_v corresponding to the imposed displacements δ_h and δ_v, respectively, were logged for post-processing purposes. Also, the vertical displacements δ_{v1} and δ_{v2} and the horizontal displacement δ_{h1} were logged at positions compatible with the locations of LVDTs in the experiment (see Fig. 29). Figure 30a shows a plot of the average vertical displacements, $\bar{\delta} = (\delta_{v1} + \delta_{v2})/2$, versus vertical load F_v and Fig. 30b shows a plot of the horizontal displacement δ_{h1} versus horizontal load F_h. The experiment curves were plotted for reference, as well as the numerical results from Nguyen (2005) and Cusatis et al. (2006). Nguyen used a non-local damage model, whereas Cusatis et al. used a confinement-shear lattice model where a set of connected particles represents the material. Both studies were able to simulate the fracture process at some

Table 13 Example 4: summary of mesh attributes

Model	Bulk elements	Cohesive elements	Nodes	DOFs	h (median) (m)	r_{min}	r_{max}
WED	1907	2822	11,458	34,374	0.0088	2.46	14.17
HEX	3396	6654	26,912	80,736	0.0071	3.40	44.05

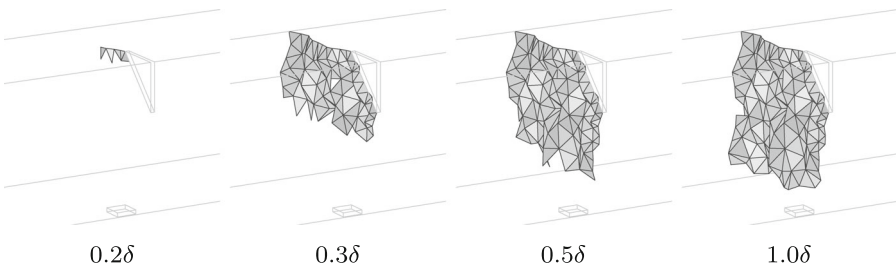


Fig. 28 Example 3: crack surface evolution for different levels of the prescribed displacement for the TET analysis

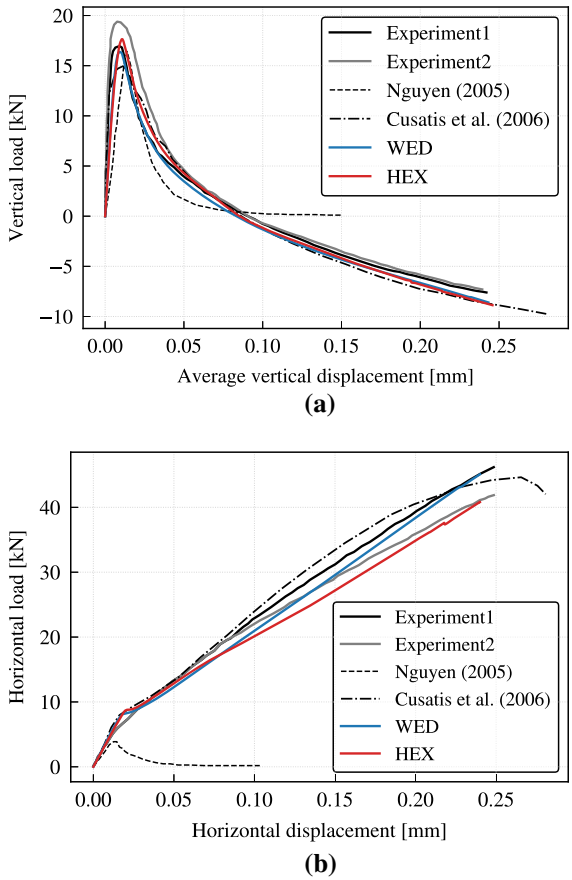


Fig. 30 Example 4: experimental and numerical results

level. However, Nguyen's results were unable to predict the rise of negative vertical force along the softening branch. Also, the horizontal load curve was not well predicted. On the other hand, the numerical curves from the current study agree with the experimental ones. The numerical curves also show similar patterns despite the differences in the type of elements and the number of degrees of freedom. Table 14 presents a comparison of the vertical peak load and corresponding displacements ($\bar{\delta}$), taking the first experiment as reference. Good agreement can be observed, especially at peak loads. Note that the discrepancies with the second experiment and the numerical results are within an order of magnitude. Regarding the elements shape, the results suggest that the proposed model can deal with elements with a high aspect ratio.

Figs. 31a and b present the deformed state, multiplied by a scale factor of 15, at the end of the two simulations. The figures also displays the τ_{xz} stress field where it is possible to see stress concentrations between the cracks as a result of the material discontinuity. Figure 31c shows the experiment crack paths for comparative purposes. Good agreement in crack patterns can be observed. Despite a finer mesh using hexahedral elements, the cracking pattern is better reproduced (the experiment crack pattern) by the mesh with wedge elements. For example, note that more than one crack appears near the left notch for the mesh with hexahedral elements. We consider that under the imposed shear conditions, interlocking prevented the crack propagation from the left notch driving more stress levels to nearby cohesive elements, which, in turn, favored the development of new cracks. For this reason and for more refined analyses, the authors suggest avoiding the use of cohesive elements combined with hexahedral elements. In addition, since this example is basically a 2D problem, and the mesh was generated by extrusion of the xz face, the obtained crack patterns are the same along the y direction. That will not be the case if, for example, unstructured meshes were used due to mesh-induced anisotropy.

6 Conclusions

This paper presents a 3D elasto-plastic constitutive model for cohesive elements to simulate fracture in quasi-brittle materials. The paper also presents some

Table 14 Example 4: comparison of numerical results and experimental data

Experiment/model	Increments	P_{max} (kN)	$\bar{\delta}$ (mm)	$\frac{\lvert\Delta P_{max}\rvert}{P_{max}}$ %	$\frac{\lvert\Delta\bar{\delta}\rvert}{\bar{\delta}}$ %
Experiment 1	–	16.96	0.0091	–	–
Experiment 2	–	19.39	0.0076	14.3	16.7
WED	992	16.38	0.0094	3.4	2.8
HEX	2090	17.65	0.0107	4.1	16.7

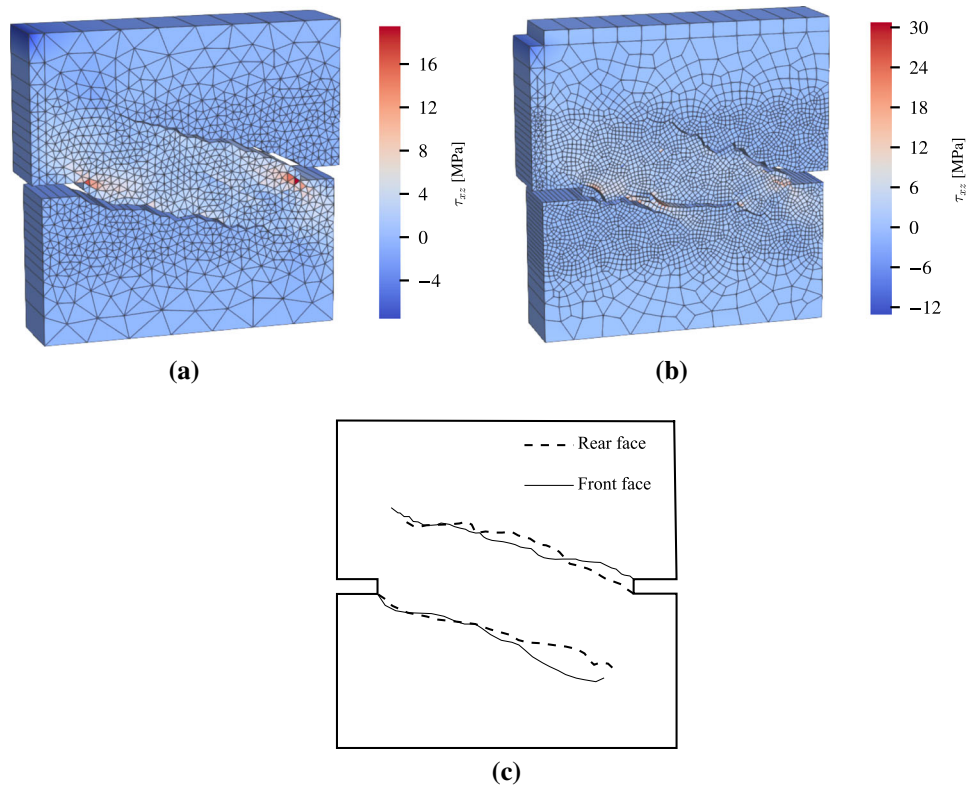


Fig. 31 Example 4: deformed meshes and experimental crack paths: **a** analysis WED; **b** analysis HEX; **c** experimental crack paths

techniques to aid the solution of the resulting nonlinear system of equations. The proposed model requires few material parameters and is able to simulate, with good and fairly good agreement, a wide range of experiments. It was tested by verifying stress components and displacements in extension and shear patch tests. Also, the results from a refinement test show little mesh dependency. This is possible thanks to variable elastic coefficients used to regulate the interface elements displacements.

Four experiments involving shear and torsion were simulated using the proposed model. In all cases, good agreement was observed in load-CMOD curves, especially when verifying peak loads and the shape of softening branches. The modeling was able to provide complete and smooth softening curves. Also, the obtained crack patterns were compatible with the experimental ones when available. Each numerical application (Examples 1 to 4) were analyzed using two different meshes. The results provided quite similar curves, demonstrating good repeatability even when working with different bulk elements types. Nonetheless, it was observed that hexahedral elements interfere with the development of cracks and, depending on the mesh refinement level, may lead to the formation of several cracks in regions where only one crack is expected. On the other hand, the use of tetrahedral and pyramidal elements tends to favor the occurrence of more realistic crack patterns.

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